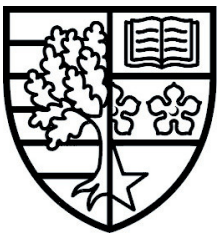
Skin Lesion Classification on FPGA

by Lewis Turner

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Dr. Rob Stewart, Ms Tess Watt



Heriot-Watt University
School of Mathematical and Computer Sciences

28th October 2026

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Course code and name	F20PA/B/C - Honours Project
Type of assessment	Individual
Document title	Skin Lesion Classification on FPGA
Student name	Lewis Turner
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• Google Gemini 3.8 Flash (Extended Thinking) to assist in searching for relevant research papers.

• Google Gemini 4.0 Argon (thinking: Low - High) to assist setting up FPGA for use over Wifi.

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Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

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1 Introduction

Summary of project motivation & context

1.1 Background and Motivation

Identifying problems to be solved to achieve objectives

1.2 Aim and Objectives

Stating project aims & objectives (enumerated)

2 Literature Review

Grouping topic summaries, critical assessment, gaps
Possibly use RADAR?

2.1 Identify

Identify key topics, themes, points

2.2 Summaries and Critical Assessments

For each summary of literature & critical assessment of topic containing:
• How already published research fits together
• Areas of consensus & disagreement
• Areas of controversy/where further research is required

2.3 Further Research Recommendations

Address an identified gap/repeat the research

2.4 Acknowledgments

Sources for literature/statements
Inserting a dummy reference for the template [1].

3 Methodology

Detailed description of necessary methods, requirements, research questions that will be employed/ needed to complete the project

3.1 Requirements

MOSCOW
Enumerated to be easier to reference again

4 Feasiblity

4.1 Preliminary Work

Summary of work carried out to prepare for this project to validate feasiblity.

4.2 Evaluation Strategy

Identify for the project:
• What needs evaulating
• Why it should be evaluated
• How it should be evaluated

5 Conclusions

Short summary of this doc

References

[1] Alan M. Turing. 2009. Computing Machinery and Intelligence. In *Parsing the Turing Test: Philosophical and Methodological Issues in the Quest for the Thinking Computer*, Robert Epstein, Gary Roberts, and Grace Beber (Eds.). Springer Netherlands, Dordrecht, 23–65. doi:10.1007/978-1-4020-6710-5_3

6 Project Management

6.1 Gantt Chart

Plan of activities to be carried out in semester 1, then future plan for semester 2

6.2 Analysis of Risks

Including mitigation plans

6.3 PLES

A well-researched consideration of any Professional, Legal, Ethical, and Social Issues pertinent to the project. (e.g. codes of conduct (BCS), codes of practice, standards, computer law, ethical decision making, intellectual property, social aspects, copyright, data protection, and so on).

7 Gen AI Summary

Summarising use of GenAI in project/deliverables